

Technology Stack

Date	04 July 2026
Team ID	SWTID-2026-4860
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution.

The **EduGenie** platform is developed using modern technologies that make the system simple, fast, and reliable. HTML and CSS provide a clean and interactive interface, FastAPI handles the backend routing operations, Google Gemini API generates intelligent educational responses. Together, these technologies help the application deliver a smooth and efficient learning experience.

Technology Stack Details

S.NO	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1.	Frontend / Client-Side	HTML5, CSS3, JavaScript	Used to build a simple and interactive interface where students enter learning queries, view structured roadmaps, and receive AI-generated quizzes.
2.	Backend / Server-Side	Python, FastAPI	Manages API requests, validates student input, coordinates AI modules, handles endpoint routing for the 5 core learning features, and returns responses efficiently.

3.	Cloud / Hosting / Deployment	Uvicorn ASGI Server (Local Host)	Used for deploying and running the application backend locally, making the portal easily accessible while supporting future scalability and maintenance.
4.	Version Control & CI/CD	Git, GitHub	Used for version control, source code management, and maintaining smooth project collaboration across the 5 team members.
5.	Third-Party APIs / Other Tools	Google Gemini API, Pydantic	Google Gemini API generates personalized explanations, summaries, and structured quizzes based on topics, while Pydantic validates data models to ensure reliable system performance.